Three-Head Convolutional Neural Network for Orientation Specific Brain Tumour Classification

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1 Introduction

In this project a fully automatic classification system is designed, meaning it requires no intervention or input other than the dataset [1], which can be applied to unseen MRI scans and is able to identify tumours no matter what orientation of brain scan is given, where the tumour is and what shape it takes. There is no current universal system for accurate classification and detection of tumours with all these factors left independent [2]. Developing a machine learning model for medical use for brain tumour classification with this dataset presents significant technical challenges, mainly due to the amount of visual heterogeneity within each tumour class. Identifying subtle distinctions between tumour types, especially considering the similarity in the appearance of Meningioma and Glioma tumours in MRI scans, requires a model capable of separating fine detailed features from the complex, anatomical background of the images.

To handle these challenges, this project compares two models, both are Convolutional Neural Networks (CNNs), these consist of an initial feature extraction section of convolutional layers , rectified linear units (ReLus) and pooling layers to reduce the dimensions of the images, as well as a classification section of fully connected neurons [3]. The first model, CNN 1, uses an initial model to classify the orientation of each scan, this is integrated into the main model by it splitting into three heads, each specialised in one orientation. This is compared to an identical CNN of just 1 'head' with no splitting for orientations, to analyse the diagnostic accuracy of them both, comparing to recent published similar experiments, and if the model can overcome orientation differences in the dataset.

2 Model Design

the two models, including and excluding the orientation classification model, both share the same base structure. Their structure diverges after the second fully connected layer, as visualised in the diagrams below.

	CNN 1 (with orientation)	CNN 2 (without orientation)
Heads	3	1
Fully Connected Layers	4	4
Diagram in Figures	Fig. 1 & 2	Fig. 1 & 3

Table 1: Comparison of the architecture of both CNNs.

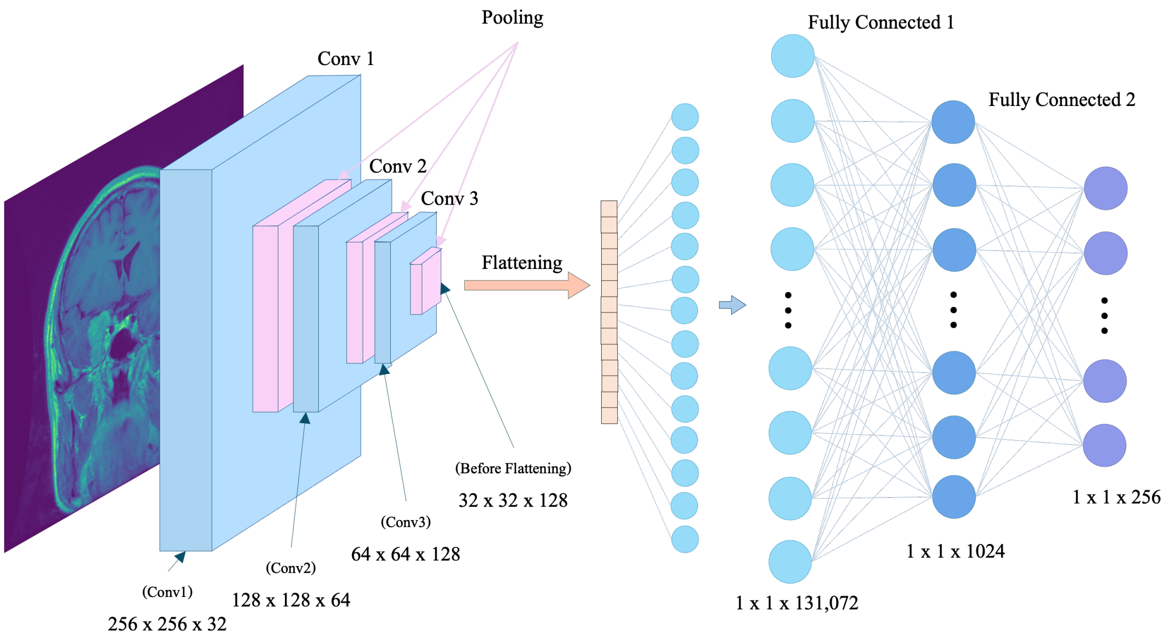

Figure 1: Diagram of the convolution neural network (CNN) designed in this project, showing the shared layers before the orientation split. This is identical for both CNN1 and 2.

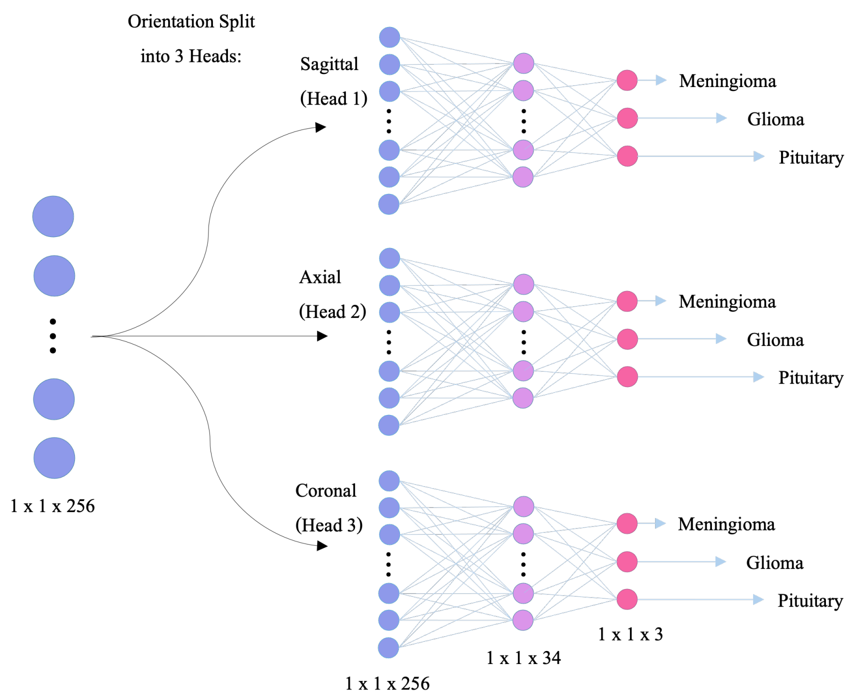

Figure 2: Continued diagram of CNN 1 (with orientation) for the 3 orientation heads, with a hidden layer and the final layer.

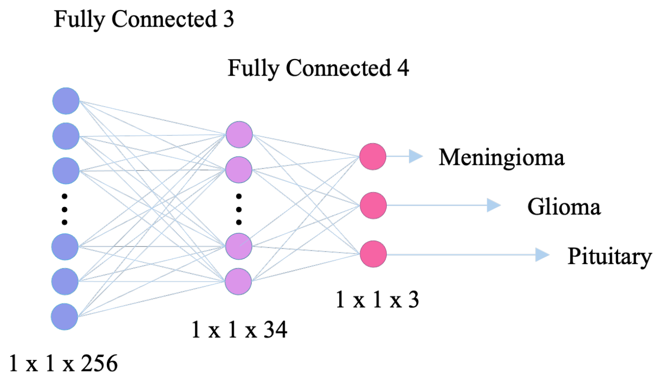

Figure 3: Continued diagram of CNN 2 (without orientation), the single 'head' model, with no inclusion of the orientation index produced by the orientation classification model.

To optimise the design of the model, the main challenge to address is how to process images of 786,432 input features (from 512×512 pixels in RGB colour) into just 3 classification probabilities. This massive reduction in dimensionality was first helped by resizing every image to (256x256) pixels, then by three convolutional layers each with a 2x2 `MaxPool2d` layer, this pooling stride was chosen to balance compressing the spacial resolution down to (32x32) after feature extraction while preserving the subtle details necessary the model must learn in order to distinguish between the similar looking Meningioma and Glioma tumours. These convolutional layers build the feature extraction section of the CNN, they perform linear transformations while preserving the spatial data [4]. Having 4 fully

connected layers which taper the reduction in dimensionality gradually down to 3. Within each fully connected layer, each neurone is 'flattened' into dimensions (1x1xlength) and connected to every other neurone, the spacial data is lost in this process [5]. For CNN 1, each image in funnelled into one of three 'heads' one for each orientation. This split was placed after the second fully connected layer to ensure enough general pattern learning first, with the final two fully connected layers after allowing enough learning of orientation specific features, ultimately increasing the overall classification accuracy of the model.

3 Training Approach

3.1 Data Preparation and Augmentation

The dataset was loaded using the `torchvision.datasets.ImageFolder` class, as this automatically indexes the images, allowing the processing of batches of images by iterable loops. To handle the images heterogeneity, a normalisation transform was put in place, shifting the pixel intensity distribution to have a mean of zero and standard deviation of 1, this prevents high intensity features from disproportionally influencing the model training. The first 5 images in each tumour type class were printed to validate they've been loaded correctly:

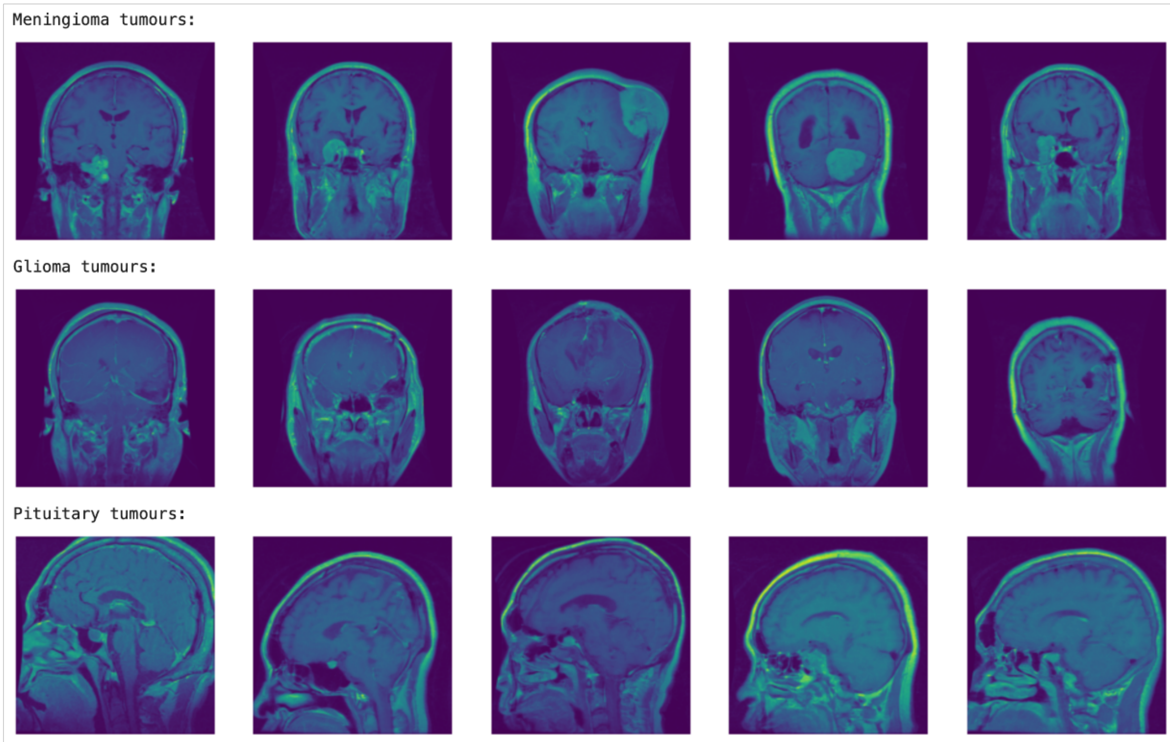

Figure 4: MRI scans from the data set, printed from the `ImageFolder` to validate they've been loaded correctly.

To mitigate the risk of overfitting, a challenge faced during the development of the code, augmentation was performed while loading the dataset to add more nuance to the dataset. A RandomRotaion of 10°was applied, a small angle was chosen to avoid disrupting the subtle patterns that model is trying to learn or distort the anatomical background as this is what the orientation model classifies with. The image size was also fixed to 256x256 pixels as mentioned.

Finally, the dataset was split into training and validation subsets in a ratio of 80 : 20, this was performed with `torch.utils.data.random_split`, ensuring a representative distribution of each tumour type in the test and train datasets. A fixed seed was used to ensure reproducibility.

3.2 Data Preparation for the Orientation model

The dataset consisted of three orientations of MRI scan: Sagittal (side on and facing to the left), Axial (from the top oriented down) and Coronal (from the back of the head). To train a model to identify orientation, a dedicated orientation specific training and testing dataset was curated by finding clusters of images of the same orientation and collecting them into subsets using `torch.utils.data.Subset` using their index. To validate the correct images are collected, their filenames and indexes were printed to cross reference.

Before loading the testing and training datasets into the model, the labels they carried had to be re-written to represent their orientation index rather than the tumour index they still carried. This was done using a designed class `LabelreWrite`. This ensured the orientation model learnt to identify orientation, not the tumour groups from which the images were taken.

3.2.1 Challenges and Optimisation

A high accuracy for the orientation model is imperative as images being mislabelled and hence training the model on conflicting information would degrade the accuracy of the overall model significantly. To prevent this, a pre trained model `torchvision.models.resnet18` was used as a starting point. This is an 18 layered model designed for image recognition, more suitable for recognising the visual patterns of the orientations than a new model. To ensure accuracy once again, the weights used were set as `default` which are predetermined from the dataset `ImageNet` containing over a million images, allowing the model to start with reasonable parameters.

Early in the design of the model, the training of the orientation classification model was unstable, occasionally producing a very low accuracy approximately 55% while often achieving the target 100% accuracy. This is due to the randomness intrinsic to the model from randomly splitting the curated dataset into test and validation subsets, as the dataset is relatively small the model is more sensitive to the shuffling of the datasets. To fix this, the test and training split was performed using a seed, making the random split deterministic allowing each run of the model to be consistently 100% accurate.

3.2.2 Validation

To validate at each step, when curating a dataset, printing images with index to ensure no misalignment. To visually validate the model, 15 images are randomly chosen from the full dataset and printed with their predicted orientation to visually check the model is performing as expected.

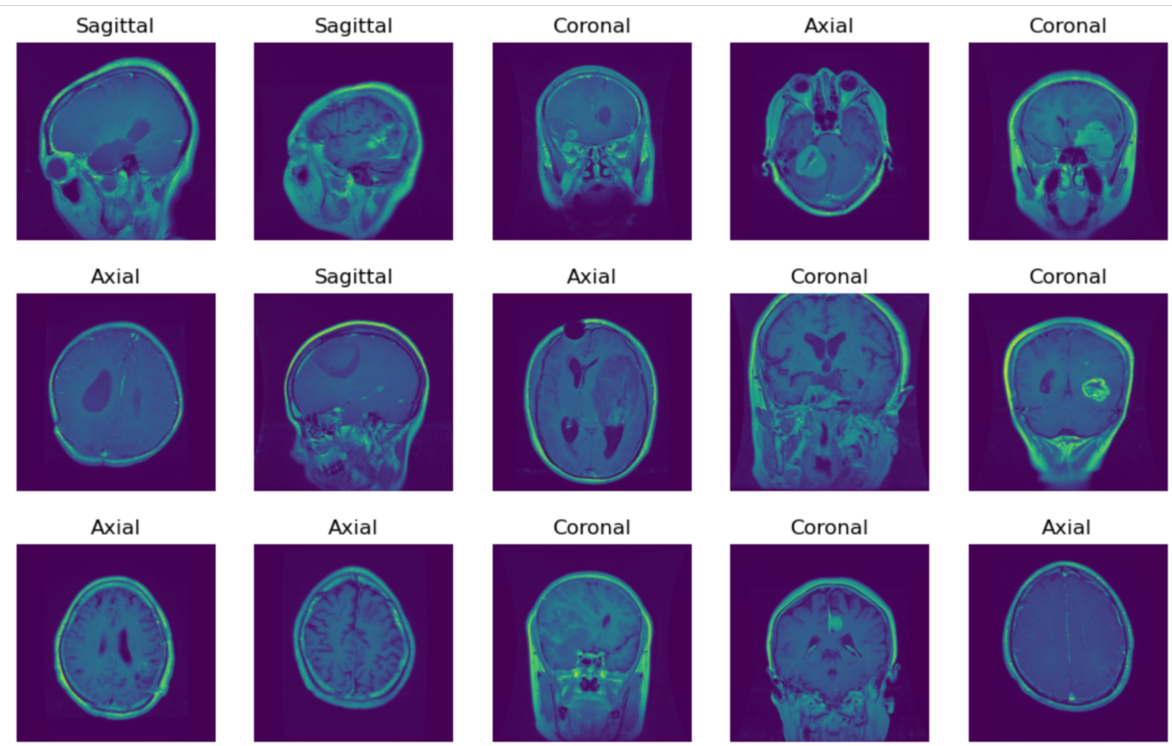

Figure 5: To validate the orientation model's high accuracy, 15 images are taken randomly from the validation dataset and printed with their predicted orientation to manually check.

3.3 Challenges and Optimisation

A challenge faced during the design of both the CNN 1 and CNN 2 model was persistent overfitting, this was shown by the validating loss plateauing and eventually increasing while training loss consistently decreased with epochs. The widening gap between training and validation loss shows the model has moved past extracting the features to identify tumours and is now memorising noise in the training dataset, reducing it's ability to be applied to unseen datasets. To counteract this, a dropout was introduced, this sets the weight of a fraction of neurones to 0 reducing the models dependency on any single neurone. A relatively low 30% dropout was layered after the first two fully connected layers, to reduce overfitting while not losing significant information by cutting off the learning between layers.

Another challenge was gradient descent instability, where the loss function fluctuates so much the model can't converge on an accurate weighting. To stabilise this, exponential learning rate decay was implemented within the Trainer class, gradually reducing learning rate to force the model to fine tune the weights.

Another strategy was using a low intensity `label_smoothing` incorporated into the `CrossEntropyLoss` function, preventing the model from aiming for 100% accuracy, allowing it to learn more generalised features. Early stopping was also used as a safety net if overfitting persists through these strategies.

3.4 Code Factorisation and Re-Usability

As the first fully connected layer `self.fc1` needs the flattened number of features the earlier 3 convolutional layers produced, a function `_get_flattened_size()` was built into the model, which calculates the number of features from the convolutional layers automatically. To further factorise the code, the Trainer class was built to be reused for both models including and excluding the 3 orientation heads, allowing the same training and exponential learning rate decay to be conduced on both. A function was designed to calculate the average confidence the model has in correct and incorrect tumour prediction, this was generalised to be applied to both CNNs.

A way the code was designed to be reusable was in the design of the class `OrientIndexAddOn`. By using this class to add model predicted orientation index onto the test and train datasets without making new ones, this ensured that both CNN 1 and CNN 2 operate with identical test and train datasets, isolating their differing architectures as the only independent variable causing their performance differences. The entire code is also reusable as it can be easily adapted for any kind of imaging classification by changing the imaging datasets and the final output layers.

3.5 Trainer Class

In training, the weights of each neurone are changed by a gradient descent to optimise the output, the function used to calculate the correctness of any output is the loss function [6], for this trainer a cross entropy loss function was chosen. This can be expressed as [7]:

$$H(y, \hat{y}) = \sum y_i \log 1/\hat{y}_i = -\sum y_i \log \hat{y}_i \tag{1}$$

Where for a class i, the expected output is $\hat{y}_i$ and the model classified output is y_i. The optimizer chosen was Adam (adaptive moment estimation) as it's designed for high noise, which this dataset has with a complex background to the tumours.

As discussed in the Challenges and Optimisation section, a learning rate decay was used to stabilise gradient descent. Initially, a higher learning rate allows the model to learn the broader features quickly, as epochs continue the scheduler tapers learning into the smaller, subtle features needed to differentiate between the tumour types, preventing the model from making any large changes at the end of the epochs that could conflict with it's earlier learning. The function chosen was $Lr = Lr_0 \cdot 0.93^{epoch}$, shown in Figure 6, so by the 25th epoch learning rate had been reduced by 83%.

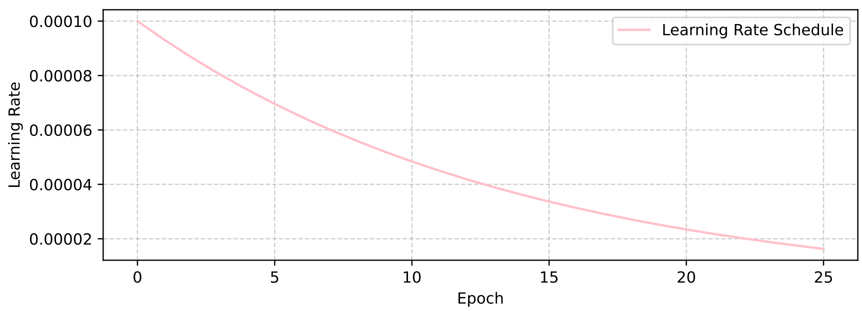

Figure 6: Planned decay of the learning rate by epoch.

To ensure overfitting couldn't put the accuracy of the final model at risk, an early stopping addition was put in place which uses weighting corresponding to the minimum validation loss, so the model is at it's highest performance.

4 Evaluation and Discussion

4.1 Orientation Classification Model

To quantitatively evaluate how well the model performed it was tested on 20% of the orientation dataset, being a total of 135 scans. The correct and incorrect model predicted orientation was collected for each image, correctly predicted orientation percentage is printed and came to 100% on the testing dataset, showing the model is very accurate. This was visually validated by printing 15 random images, as mentioned.

	Meningioma	Glioma	Pituitary	Total
Sagittal	269	325	433	35.5%
Axial	208	396	362	32.4%
Coronal	231	705	135	32.1%

Table 2: Number of MRI scans identified as each orientation, per tumour type.

To visualise the results the number of each orientation of scan within the 3 folders of tumour type was summed up and printed. To present this data in a more easily interpretable way, a matrix was made with each row representing 100% of each tumour type's brain scans, with each column representing the proportion which are identified by the model as a certain orientation.

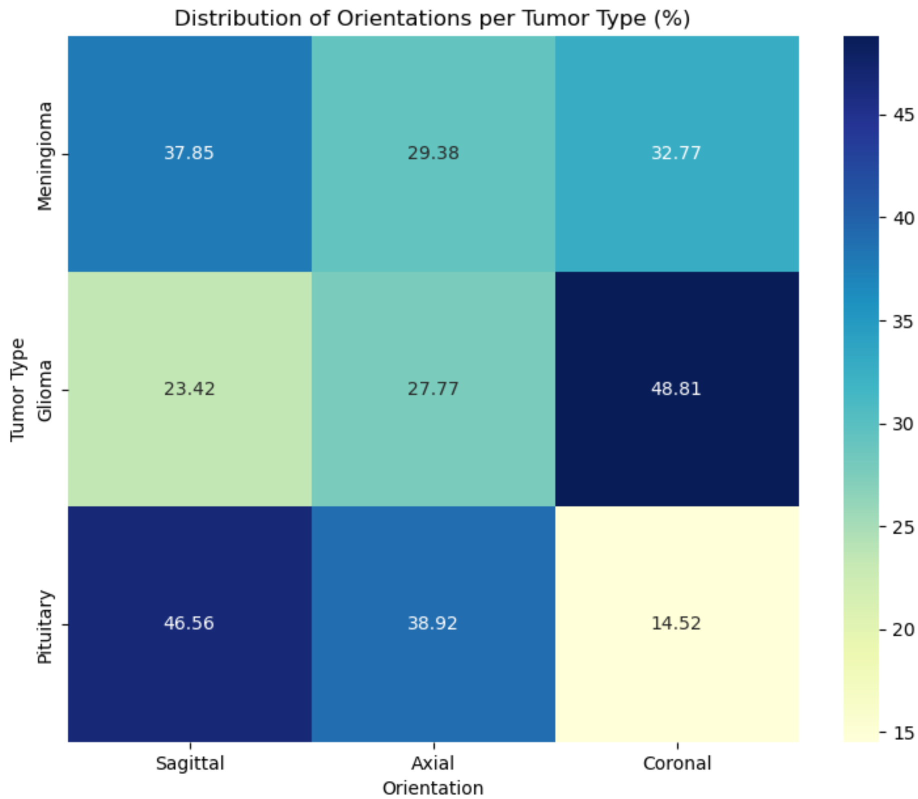

Figure 7: Heat Map of orientation of scan by tumour class.

4.2 CNN 1

To visualise learning during training, training and validation loss is plotted with epoch:

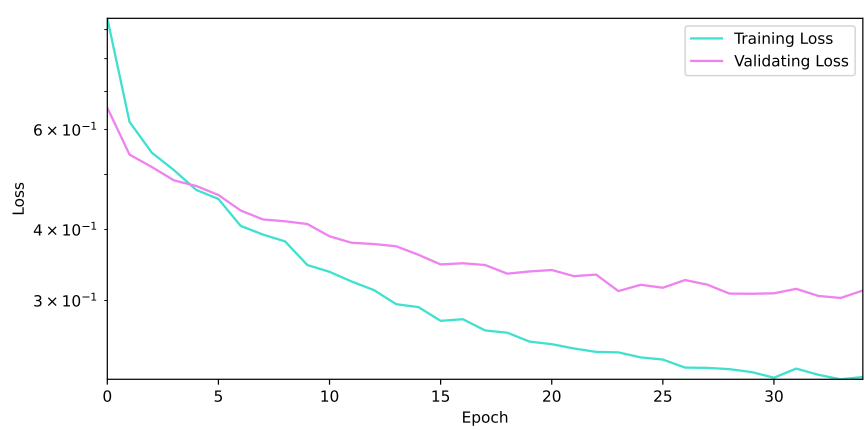

Figure 8: Training and Validation loss with epoch.

Figure 8 shows a consistent decline in both training and validation loss, showing that the model is successfully identifying features and patterns from the MRI scans and improving in it's recognition of tumour types. However, with the widening generalisation gap and the eventual plateau of the validation loss while training loss continues to decrease, this suggests the beginning of overfitting despite the several measures put in place to reduce this.

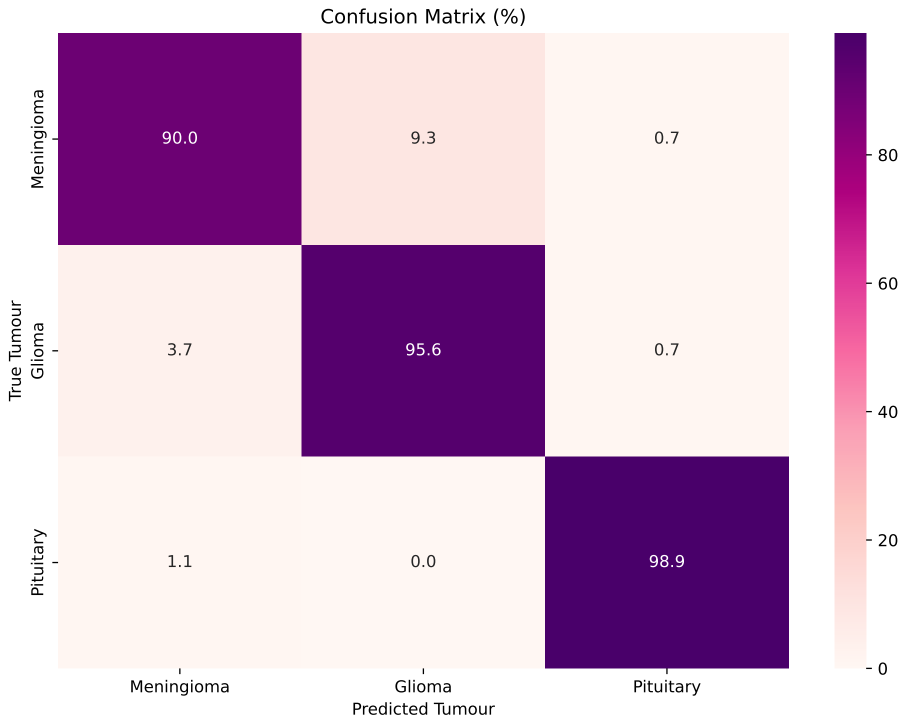

Figure 9: True Tumour type with model predicted Tumour type matrix.

An overall accuracy rate was calculated as a mean of the trace of Figure 9, which came to 95.24%. The majority of false predictions were between the Meningioma and Glioma tumours, this was expected due to their visual similarity.

The average confidence was calculated for the true and false model predictions:

Confidence	CNN 1
True Predictions	93.48%
False predictions	69.92%

Table 3: Average confidence of the model for it's true and false tumour predictions for CNN 1.

The high confidence in true predictions of 93.48% shows the model has internal confidences aligning with it's high accuracy, as well as showing the learning rate decay has successfully stabilised the gradient descent. The gap in confidence for true and false prediction for CNN 1 came to 23.56%, demonstrating the model isn't confidently wrong in more difficult cases.

4.3 CNN 2

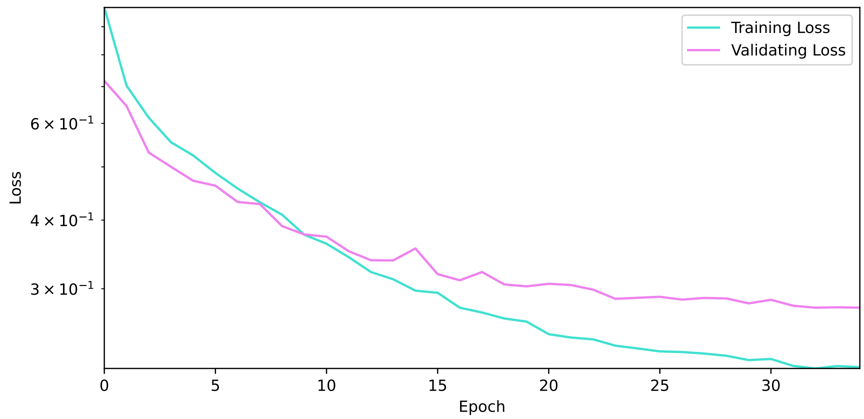

Figure 10: Training and Validation Loss with Epoch for CNN 2.

Figure 10 shows a similar story to figure 8, which is expected. However the generalisation gap is smaller and doesn't increase with epoch at as high of a rate, this suggests less overfitting. CNN 1 shows more signs of overfitting to CNN 2, shown by figures 8 and 10. This could be due to CNN 1 having three heads, each therefore having a smaller subset to train each head on could more easily lead to memorisation.

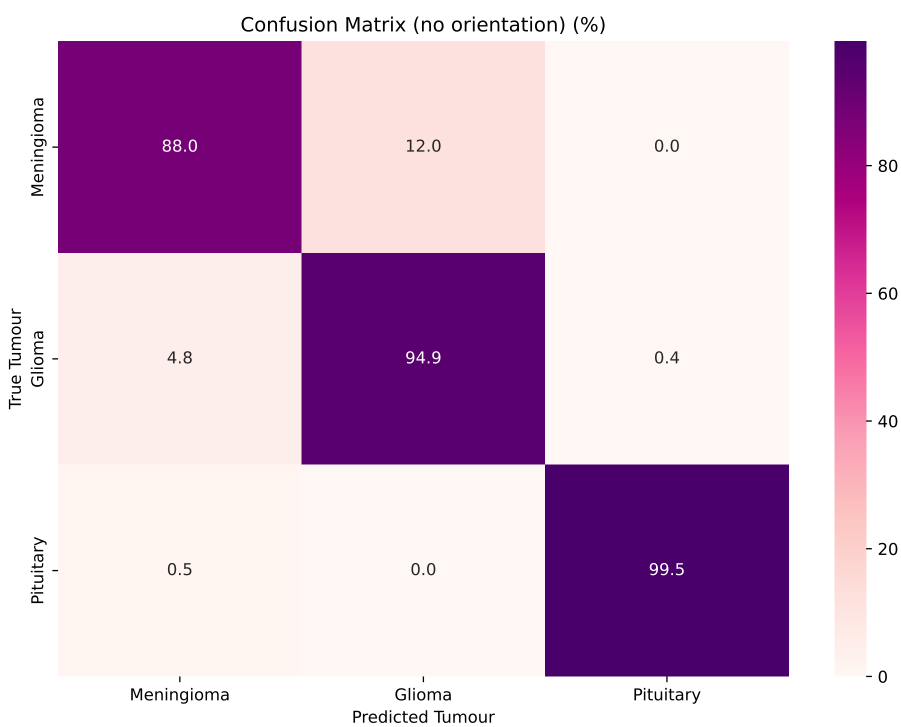

Figure 11: True Tumour type with model predicted Tumour type matrix.

The overall accuracy rate if CNN 2 was 94.62%. This is expected as CNN 2 is less accurate overall and so will intensify the inaccuracies already present in CNN 1. This could be improved by increasing the size of the dataset and ensuring even orientation distribution between tumour types.

Confidence	CNN 2
True Predictions	86.90%
False predictions	36.01%

Table 4: Average confidence of the model for it's true and false tumour predictions for CNN 2.

4.4 Comparison

True prediction Rate (%)	CNN 1	CNN 2	Difference
Meningioma	90.0	88.0	+2.0%
Glioma	95.6	94.9	+0.7%
Pituitary	98.9	99.5	-0.6%
Overall	95.24	94.62	+0.62%

Table 5: Comparison of true prediction rate results of both CNNs, from Figure 9 and 10.

The highest overall accuracy was from CNN 1, higher by 0.62%, this was most prominent in the classification of Meningioma tumours. which was 2% increased from the addition of the 3 heads structure, demonstrating it causes more detailed processing of the scans within orientations leading to higher classification accuracy.

Confidence (%)	CNN 1	CNN 2	Difference
True Predictions	93.48	86.90	+6.58%
False predictions	69.92	36.01	+33.91%
Gap	23.56	50.89	

Table 6: Comparison of the results of both CNNs.

The confidence analysis for CNN 2 showed less certainty for correct predictions compared to CNN 1, reinforcing the conclusion that the inclusion of orientation in CNN 1 improved the model's ability to converge and become more decisive. Overall, Meningioma were the most difficult to classify. This

is due to their similarity visually to Pituitary tumours and is exacerbated by their disproportionally small dataset, accounting for 23% of the full dataset. To improve both models, a evenly distributed dataset would allow the model to train evenly on each tumours, reducing the off-diagonals in figures 9 and 11. To improve the model, images of brain scans of each orientation without any tumours would

be beneficial as to not muddy the visual patterns the model has to learn from, however as the tumours have such little visual input in comparison to orientation in each image this is unlikely to have a significant effect on the model's learning.

5 Conclusion

In conclusion, two CNNs were designed to identify three tumours without any prior processing. The highest accuracy and certainty was from using an orientation classification model integrated into the CNN after feature extraction and two fully connected layers, this had an overall accuracy of 95.24%, which outperformed other results using CNNS for this purpose, which had a validation accuracy of 84.19% [8]. To improve, a more evenly distributed tumour types dataset would be desirable as well as a new class of healthy brain scans to extend the models capabilities to not just classifying tumours but identifying the presence of one.

6 Use of Generative AI

- Asked Google Gemini for a suggestion for a pre trained model to use for the orientation classification model, it recommended the `torchvision.models.resnet18` which I used.
- Used Google Gemini for general help in the use and training of the pre trained model resnet18.
- Asked Google Gemini for how to set a seed to make random splitting reproducable, it gave lines `torch.backends.cudnn.deterministic = True` and `torch.backends.cudnn.benchmark = False`.
- Used Google Gemini to clarify general code errors such as IndexError, FloatingPointError and TypeError.
- Used Google Gemini for help in the understanding of classes and instances.

References

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[8] Abiwinanda, Nyoman Abiwinanda, Muhammad Hanif, S. Tafwida Hesaputra, Astri Handayani, and Tati Rajab Mengko. *Brain Tumor Classification Using Convolutional Neural Network.* World Congress on Medical Physics and Biomedical Engineering. (2019).